Test report 13 – geometry fixes

Geometry and kernel bugs from the everyday models (`examples/alltag/README.md`, `tests/alltag/fehler/`) · SMLib facade, SMLib patches 0002/0003, cad-engine, cad-render, cad-cli · 2026-10-23 · Branch worktree-agent-a5315caa6a497a5f8 · Commit e04a998 · 4 cores

cargo test --workspace
426 / 426
of which 22 new regression tests

clippy · fmt
clean · clean
-D warnings

Wasm kernel (Node)
passed
facade rebuilt

OpenSCAD · STEP
33/33 · 48/48 passed
compare.py, verify.py

Everyday models
19 models, 0 with problems
volumes unchanged

Bug examples
9 fixed, 8 open
against their expected state

Key rack
16.7 → 6.1 s
evaluation per call

1 Overview

The everyday models found one crash, four silently wrong results and a number of kernel bugs. Each minimal example in `tests/alltag/fehler/` now carries its expected state in its header (`// SOLL: fehlerfrei` = error-free, `SOLL-VOLUMEN` = expected volume, `SOLL-FEHLER: Text` = expected error – where there is no correct result, a clear message is the goal); `tests/alltag/check.sh --fehler` reports „behoben“ (fixed) or „offen“ (open) per example. The table shows both states (before: state before the fixes, commit `cba3ab4`; after: this commit), `cad measure --volume --json`.

Example	Expected	before	after	check.sh
`f01_sweep_spline_profil`	SOLL: fehlerfrei	V 0.0 mm³ 1 Fehler: sweep gescheitert: not supported by this kernel: sweep of a profile that is not centred on its origin along a curved path	V 2,179.6 mm³ no message	fixed
`f02_union_zylinder_sweep`	SOLL: fehlerfrei	V 51,429.5 mm³ 1 Fehler: Vereinigung gescheitert: boolean union failed: SM_ERR_NULL_POINTER (1006)	V 51,429.5 mm³ 1 Fehler: Vereinigung gescheitert: boolean union failed: SM_ERR_NULL_POINTER (1006)	open
`f03_sweep_polygon16`	SOLL: fehlerfrei	V 0.0 mm³ 1 Fehler: sweep gescheitert: sweep did not produce a closed solid (self-intersecting or open profile?)	V 0.0 mm³ 1 Fehler: sweep gescheitert: sweep: SMLib left 3 seam(s) between the side faces of the polygon profile open (SMLib limitation on curved paths; circle() or a spl	open
`f04_union_fillet_sweep_polygon`	SOLL: fehlerfrei	V 1,237.4 mm³ 1 Fehler: Verrundung gescheitert (1 mm): fillet failed: SM_ERR (1001)	V 1,237.4 mm³ 1 Fehler: Verrundung gescheitert (1 mm): fillet failed: SM_ERR (1001)	open
`f05_fase_volumen_null`	SOLL-FEHLER: not closed	V 0.0 mm³ no message	V 20,102.9 mm³ 1 Fehler: Fase gescheitert (0.8 mm): chamfer failed: SMLib returned a body that is not closed (faces missing): the blends of neighbouring edges collide, size to	fixed

f06_fase_konkave_ecke	SOLL: fehlerfrei	V 26,174.0 mm³ 1 Fehler: Fase gescheitert (0.5 mm): chamfer failed: SM_ERR (1001)	V 26,174.0 mm³ 1 Fehler: Fase gescheitert (0.5 mm): chamfer failed: SM_ERR (1001)	open
f07_torus_zone	SOLL: keine Warnungen	V 12,566.4 mm³ warning: 0 edges	V 12,539.7 mm³ no message	fixed
f08a_union_torus	SOLL-VOLUMEN: 5852.26 1	V 6,049.7 mm³ no message	V 5,852.3 mm³ no message	fixed
f08b_union_ring_verloren	SOLL-VOLUMEN: 5852.26 1	V 5,654.9 mm³ no message	V 5,852.3 mm³ no message	fixed
f08c_differenz_torus_tangential	SOLL-FEHLER: not a closed solid	V 0.0 mm³ no message	V 5,654.9 mm³ 1 Fehler: Differenz gescheitert: the result is not a closed solid: the operands touch tangentially or along an edge there (let them overlap a little or keep a g	fixed
f09a_absturz_tangentialer_dom	SOLL-FEHLER: touch tangentially	crash (exit 139), no output	V 1,584.8 mm³ 1 Fehler: Vereinigung gescheitert: the result is not a closed solid: the operands touch tangentially or along an edge there (let them overlap a little or keep a	fixed
f09b_absturz_zapfen_sweep	SOLL-FEHLER: cannot blend the corner	crash (exit 139), no output	V 1,067.1 mm³ 1 Fehler: Verrundung gescheitert (1 mm): fillet failed: SMLib cannot blend the corner where these edges meet (faces that touch tangentially there?)	fixed
f10_union_fillet_schale_verloren	SOLL: fehlerfrei; SOLL-VOLUMEN-MIN: 20000	V 0.0 mm³ no message	V 26,042.9 mm³ 1 Fehler: Verrundung gescheitert (1 mm): fillet failed: SMLib returned a body that is not closed (faces missing): the blends of neighbouring edges collide, size	open
f11_stift_deckungsgleich	SOLL: fehlerfrei	V 10,165.1 mm³ 1 Fehler: Vereinigung gescheitert: boolean union failed: SM_ERR_NULL_POINTER (1006)	V 10,165.1 mm³ 1 Fehler: Vereinigung gescheitert: boolean union failed: SM_ERR_NULL_POINTER (1006)	open
f12_shell_loft_quadrat_kreis	SOLL: fehlerfrei	V 515,686.3 mm³ 1 Fehler: shell gescheitert (2 mm): SmApiShellBrepFull failed: SM_ERR_NULL_POINTER (1006); kernel trace: SM_ERR_NULL_POINTER at SmMerge.cpp:672; shell from bool	V 515,686.3 mm³ 1 Fehler: shell gescheitert (2 mm): SmApiShellBrepFull failed: SM_ERR_NULL_POINTER (1006); kernel trace: SM_ERR_NULL_POINTER at SmMerge.cpp:672; shell from bool	open
f13_shell_verrundung	SOLL-FEHLER: radius of a convex rounding	V 59,848.6 mm³ 1 Fehler: shell gescheitert (2 mm): SmApiShellBrepFull failed: SM_ERR (1001); shell from booleans: SmApiOffsetBrepFull failed: SM_ERR (1001)	V 59,848.6 mm³ 1 Fehler: shell gescheitert (2 mm): shell: wall thickness 2.000000 is not smaller than the radius of a convex rounding of the body (about 2.000000): the inner s	fixed
f15_knauf_kehle	SOLL: fehlerfrei	V 1,118.4 mm³ 1 Fehler: Verrundung gescheitert (1 mm): fillet failed: SM_ERR (1001)	V 1,118.5 mm³ 1 Fehler: Verrundung gescheitert (1 mm): fillet failed: SM_ERR (1001)	open

Nothing silently wrong any more: of the findings with a wrong result and no message (F5, F8a–c, F10), F8a/F8b are now computed correctly, F5, F8c and F10 are reported (F5 and F8c have no correct result: colliding chamfers, tangential contact). No example crashes any more.

2 Fixes in the facade and in SMLib

Location	Change	Findings
`smlib/patches/0002-fillet-missing-corner.patch`	SmFilletExecutive::CreateFilletCorners: vertex without corner type → error instead of type confusion (crash)	F9
`smlib/patches/0003-sweep-profile-reference-point.patch`	SmSweepOptions::bUseProfileRefPoint/sProfileRefPoint: specify the reference point of the profile for curve sweeps	F1
`apply_matrix`	Transform(m, FALSE): rigid motions keep the tolerances, scaling sets them from the whole bounding box; context untouched	F7, F8a, F8b
`checked_boolean`	non-closed result of two closed bodies or impossible volume → error instead of "last resort"; merely touching bodies: a − b = a, a ∩ b = empty	F8c, F9a
`blend_core`, `do_fillet_like`	result of a fillet/chamfer: closed, volume plausible; separate edge chains one after another; message for a missing corner type	F5, F10, F9b
`sc_sweep`	profile origin as reference point (0003); open side seams of polygon profiles reported with cause	F1, F3
`sc_shell`	message with the smallest convex radius ≤ wall thickness	F13
`sc_mass_props`	volume with relative accuracy 1e-6 instead of 1e-8 (error ≤ 5e-6 measured)	run time

The patches are in `smlib/patches/`; `smlib/fetch.sh` applies them in order. Native libraries: `smlib/build-native.sh`; the Wasm kernel is rebuilt (`smlib/build-wasm.sh`) and passes `node crates/kernel-smlib/js/test-node.mjs`.

3 Findings in detail

F9 – crash with union(fillet) and tangential contact

Examples `tests/alltag/fehler/f09a_absturz_tangentialer_dom.scad`, `tests/alltag/fehler/f09b_absturz_zapfen_sweep.scad`

Cause In `SmFilletExecutive::CreateFilletCorners` (SMLib) a corner type is chosen for each vertex of the edges to be filleted. If there is no suitable one (`SmFilletCorner::Create` returns NULL, "Unimplemented corner case"), the vertex was skipped with `continue` – but the table `m_vVertexCornerMap` still held the list of its edges for it (`SmTArray<SmEdgeuse>`). `GetFilletCornerOfVertex` later returned this list as `SmFilletCorner*`; `CollectTangentEdgeSequence` called virtual methods on it → access to address 0x2f08 (gdb with debug objects, valgrind: "not stack'd, malloc'd or free'd"). In F9a the missing corner type is the tangential contact boss/wall, in F9b the nearly tangential circular edge pin/sweep.

Fix **smlib/patches/0002-fillet-missing-corner.patch:** without a corner type the entry is removed and the fillet ends with `SM_ERR_INVALID_INPUT`, the edge is recorded in `SmFilletErrorInfo`. The facade translates this into "SMLib cannot blend the corner where these edges meet (faces that touch tangentially there?)" with the edge (red in `render --line`). In F9a the union already reports the tangential contact (see F8c): boss and wall touch along a line, there is no manifold result. The daemon fallback was no solution: the crash comes from the process itself.

Evidence `f09a_absturz_tangentialer_dom`: before **crash (exit 139), no output** → after V 1,584.8 mm³

1 Fehler: Vereinigung gescheitert: the result is not a closed solid: the operands touch tangentially or along an edge there (let them overlap a little or keep a

`f09b_absturz_zapfen_sweep`: before **crash (exit 139), no output** → after V 1,067.1 mm³

1 Fehler: Verrundung gescheitert (1 mm): fillet failed: SMLib cannot blend the corner where these edges meet (faces that touch tangentially there?)

Tests: `f09_examples_report_errors_instead_of_crashing` **ok**,

`f09_tangent_contact_under_union_fillet_fails_cleanly` ok,
`fillet_at_a_nearly_tangent_circle_fails_without_crash` ok

F8 – booleans with torus and rotate_extrude

Examples `tests/alltag/fehler/f08a_union_torus.scad`, `tests/alltag/fehler/f08b_union_ring_verloren.scad`, `tests/alltag/fehler/f08c_differenz_torus_tangential.scad`

Cause Not the torus itself but its **transformation**: the facade called `SmBrep::Transform(m, TRUE)`; with `TRUE` SMLib derives the model size from the bounding box of the *vertices* and sets all tolerances (vertices, edges, faces, and those of the global context) from it. A torus and the solid of revolution of a circle have exactly one vertex – bounding box 0, tolerance 0. With tolerance 0 the surface intersection misses every intersection curve: union = both bodies side by side (F8a, 6049.65 mm³ = sum), ring lost (F8b), intersection empty. Verified: the same torus created directly at z = 2 intersects correctly, moved (or only rotated or scaled) it does not; the geometry (B-rep stream) is identical, only the tolerances 1e-5 → 0 differ. F7 (torus as zone selects no edge) has the same cause.

Fix `apply_matrix` transforms with `Transform(m, FALSE)`: rigid motions leave the tolerances unchanged; with scaling the facade sets them from the whole (loose) bounding box of the body, the context stays untouched. F8c: the torus touches top and bottom face tangentially along two circles; the result is not manifold (SMLib returns it with the correct volume 5260.36 mm³, but not closed; `measure` showed 0). The boolean check no longer accepts a non-closed result of two closed bodies as a "last resort" but reports "the result is not a closed solid: the operands touch tangentially or along an edge there (let them overlap a little or keep a gap)". If the bodies only touch (interiors do not overlap), a − b = a and a ∩ b = empty apply instead of a non-manifold body.

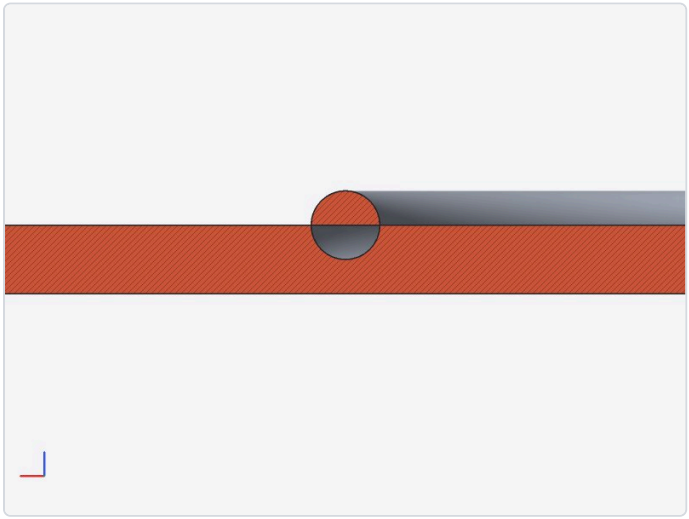
Evidence

`f08a_union_torus`: before V 6,049.7 mm³
no message → after V 5,852.3 mm³
no message

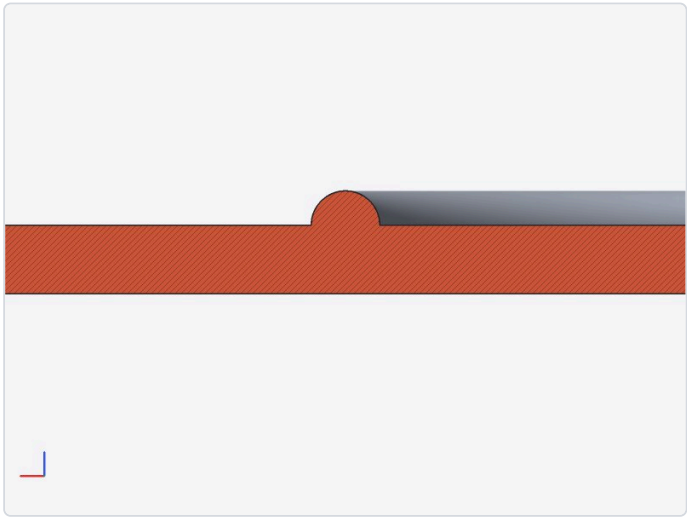
`f08b_union_ring_verloren`: before V 5,654.9 mm³
no message → after V 5,852.3 mm³
no message

`f08c_differenz_torus_tangential`: before V 0.0 mm³
no message → after V 5,654.9 mm³
1 Fehler: Differenz gescheitert: the result is not a closed solid: the operands touch tangentially or along an edge there (let them overlap a little or keep a g

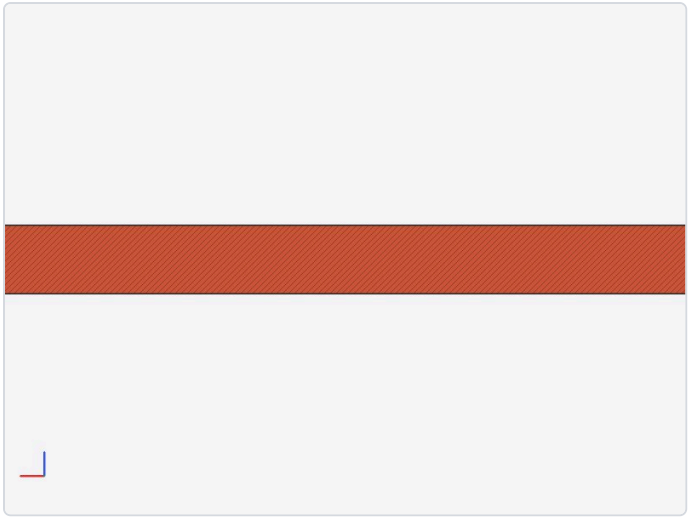
Tests: `f07_translated_torus_selects_the_edge_on_its_core_circle` ok, `f08_booleans_with_moved_tori_and_revolved_circles` ok, `moved_revolved_ring_unites` ok, `tangent_torus_in_a_slab_is_an_error` ok, `difference_of_bodies_touching_along_an_edge_keeps_the_first` ok, `moved_rotated_and_scaled_tori_intersect` ok



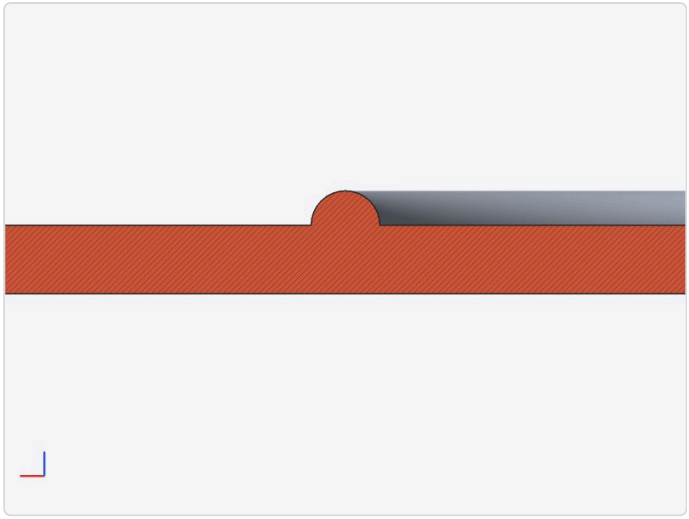
Before: two overlapping bodies, no intersection with the top face



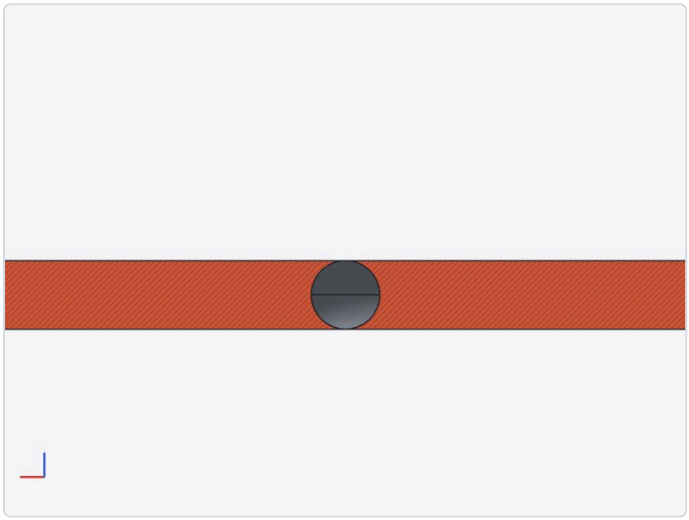
After: one body, the ring half inside the plate



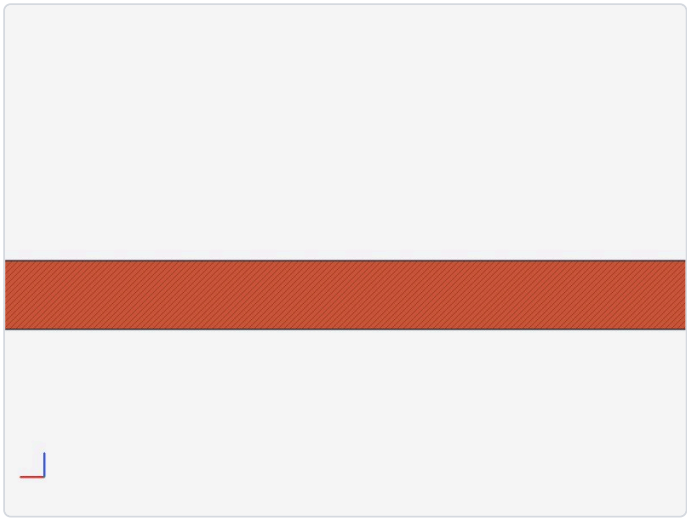
Before: ring missing



After: ring present



Before: volume 0 without message



After: error message, plate remains

F5 – chamfer via zone on the filleted box: volume 0, status ok

Examples `tests/alltag/fehler/f05_fase_volumen_null.scad`

Cause SMLib returns `SM_SUCCESS` and an open shell: the top face is missing. The reason is the geometry: at the corners the wall between outer radius 6 and the sharp inner corner is only 6 − 3√2 ≈ 1.76 mm thick; the outer chamfer takes 0.8 mm, the inner chamfer at the sharp corner 0.8·√2 ≈ 1.13 mm – the chamfers intersect, the top face falls apart. Re-measured: `chamfer(0.7)` works, `chamfer(0.75)` does not (computed limit 0.73). Chamfers of the outer and of the inner edge each work on their own.

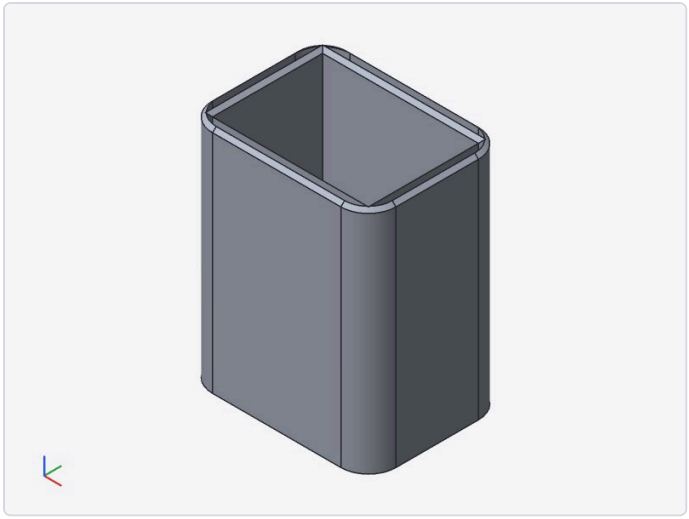
Fix The facade checks every result of a fillet/chamfer (`blend_core`): closed shell if the input was closed; volume > 0 and volume change at most as large as a fillet of this size can cause near the edges (20 r² per mm of edge length + 10 r³ per edge + 2 %). Violated → "SMLib returned a body that is not closed (faces missing): the blends of neighbouring edges collide, size too large for the wall between them?" with the edges. In addition: if the selection fails as a whole and splits into separate edge chains (without a shared vertex), the chains are filleted one after another (edges found again in the intermediate body via their geometry) – futile here because the chamfers really collide, but helpful where SMLib only fails on the combination.

Evidence `f05_fase_volumen_null`: before V 0.0 mm³

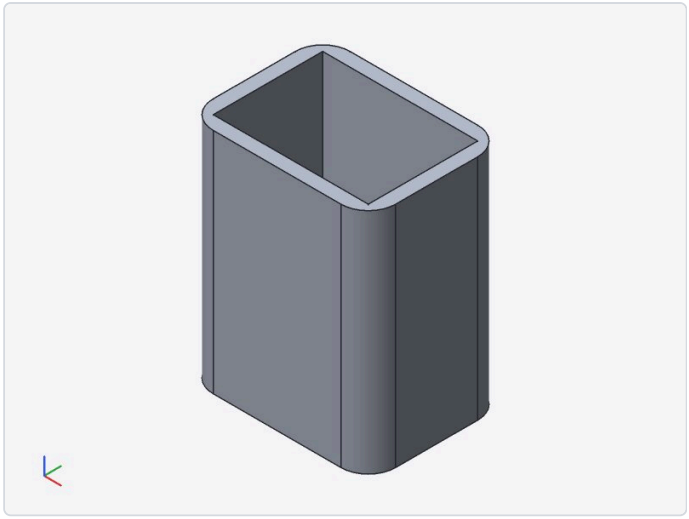
no message → after V 20,102.9 mm³

1 Fehler: Fase gescheitert (0.8 mm): chamfer failed: SMLib returned a body that is not closed (faces missing): the blends of neighbouring edges collide, size to

Tests: `f05_chamfer_that_leaves_an_open_shell_is_an_error` **ok**, `chamfer_result_must_be_closed` **ok**



Before: open shell (26 faces), volume 0, no message



After: error message, body without chamfer remains

F10 – union(fillet) of filleted shell and boss: only the boss

Examples `tests/alltag/fehler/f10_union_fillet_schale_verloren.scad`

Cause The union itself is correct (26,043 mm³). The concave fillet r 1 around the boss (7.5 mm from the inner corner) runs onto the bottom fillet of the shell (inner radius 1.5 between x = 2 and 3.5); SMLib then returns an open body consisting of boss and fillet.

Fix The plausibility check of the fillet (F5) detects the open shell: error with the edges, the union without the concave fillet remains (instead of volume 0 without message). A correctly computed fillet across two faces remains open (SMLib).

Evidence `f10_union_fillet_schale_verloren`: before V 0.0 mm³

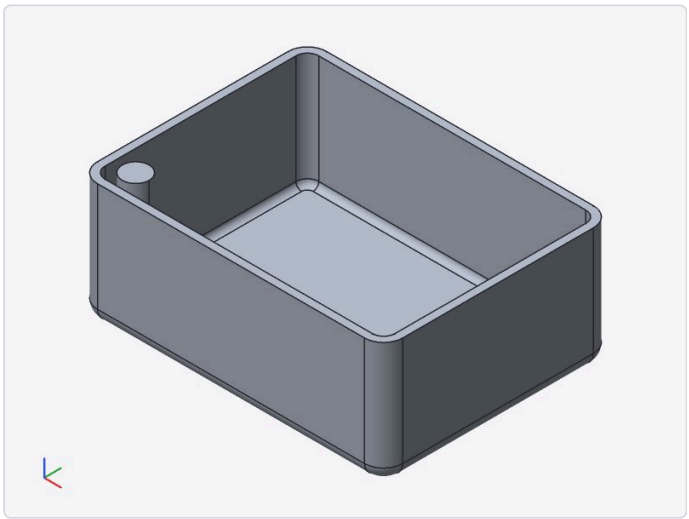
no message → after V 26,042.9 mm³

1 Fehler: Verrundung gescheitert (1 mm): fillet failed: SMLib returned a body that is not closed (faces missing): the blends of neighbouring edges collide, size

Tests: `f10_union_fillet_that_loses_the_shell_is_an_error` **ok**



Before: only boss with fillet, volume 0



After: error message, union without fillet remains

F1 – sweep with symmetric spline profile "not centred"

Examples `tests/alltag/fehler/f01_sweep_spline_profil.scad`

Cause SMLib (`CreateCurveSweep`, `bMoveProfileCenterToPath`) places the center of the bounding box of the profile curves on the path. `CalculateBoundingBox` of a B-spline curve is the bounding box of its *control points*: for the closed interpolating spline through (5,0) (0,3) (−5,0) (0,−3) x −15…5, although the curve itself is symmetric (sampled ±5, ±3.08). The facade rejected every such "off-center" profile on a curved path.

Fix **smlib/patches/0003-sweep-profile-reference-point.patch:** `SmSweepOptions` gets `bUseProfileRefPoint`/`sProfileRefPoint`; the facade specifies the origin of the profile frame (path start). Any profile, including deliberately off-center ones (circle at (0, 4)), thus follows the path with its origin; the shift after the sweep on straight paths is no longer needed. Only the mitered variant for planar polyline paths lacks the option (the message names the workaround: spline path).

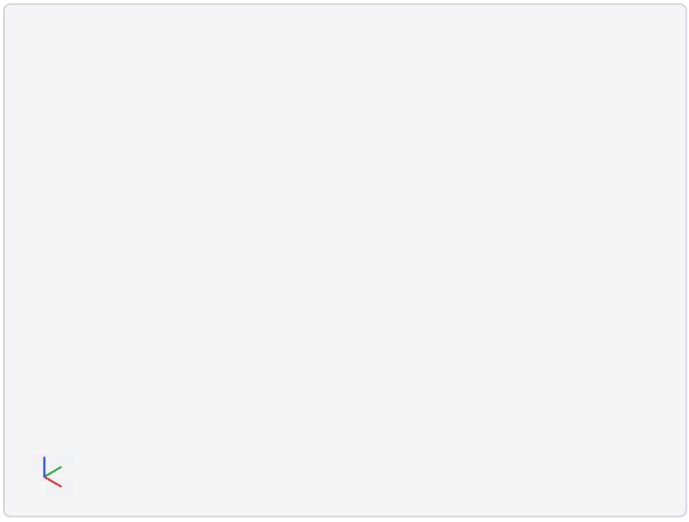
Evidence `f01_sweep_spline_profil`: before V 0.0 mm³

1 Fehler: sweep gescheitert: not supported by this kernel: sweep of a profile that is not centred on its origin along a curved path

→ after V 2,179.6 mm³

no message

Tests: `f01_symmetric_spline_profile_sweeps_along_a_curved_path` **ok**, `f01_off_centre_profile_follows_its_origin_on_a_curved_path` **ok**, `sweep_of_off_centre_and_spline_profiles_along_a_curve` **ok**



Before: rejected (no body)



After: sweep of the ellipse

F2 – cylinder ∪ curved sweep: NULL_POINTER, duplicate diagnostic

Examples `tests/alltag/fehler/f02_union_zylinder_sweep.scad`

Cause The sweep is valid (closed, no overlapping faces) and unites with cuboids, even when they cut the end cap. With the cylinder (also as NURBS, also rotated) SMLib's surface intersection fails near the tilted end cap (cap x 16.3…21.7 intersects r = 20). The facade tries orderings and offsets of 1e-6…1e-4 – without success. The duplicate message came from the engine: the failed union returns its operands, the root unites them again and fails a second time.

Fix Engine: pieces from a failed boolean operation carry a marker (`NodeOutput::unmerged`); if the union in the parent node fails on them again, this is not reported a second time. The kernel bug itself remains open.

Evidence `f02_union_zylinder_sweep`: before V 51,429.5 mm³

1 Fehler: Vereinigung gescheitert: boolean union failed: SM_ERR_NULL_POINTER (1006) → after V 51,429.5 mm³

1 Fehler: Vereinigung gescheitert: boolean union failed: SM_ERR_NULL_POINTER (1006)

Tests: `f02_failed_union_is_reported_once` **ok**

F3, F4 – sweep of a 16-gon "not a closed solid"; union(fillet) on a 32-gon sweep

Examples `tests/alltag/fehler/f03_sweep_polygon16.scad`, `tests/alltag/fehler/f04_union_fillet_sweep_polygon.scad`

Cause SMLib builds one side face per polygon side; between some neighbors, twin edges at distance 0 stay unstitched (1 face each). `StitchFaces`/`SmApiStitchAdvanced` with tolerance 1e-5…1e-3 do not stitch them either (0 stitched). On other paths 12- or 32-gons are affected. F4: the fillet of the 32 facet edges at the circular edge fails (`SM_ERR`).

Fix Message with cause: "SMLib left 3 seam(s) between the side faces of the polygon profile open (SMLib limitation on curved paths; circle() or a spline() profile work)". A fix in SMLib (construction of the side faces in the sweep) remains open.

Evidence

`f03_sweep_polygon16`: before V 0.0 mm³

1 Fehler: sweep gescheitert: sweep did not produce a closed solid (self-intersecting or open profile?) → after V 0.0 mm³

1 Fehler: sweep gescheitert: sweep: SMLib left 3 seam(s) between the side faces of the polygon profile open (SMLib limitation on curved paths; circle() or a spl

`f04_union_fillet_sweep_polygon`: before V 1,237.4 mm³

1 Fehler: Verrundung gescheitert (1 mm): fillet failed: SM_ERR (1001) → after V 1,237.4 mm³

1 Fehler: Verrundung gescheitert (1 mm): fillet failed: SM_ERR (1001)

Tests: `f03_open_polygon_sweep_names_the_cause` **ok**

F13 – message for shell of a body with fillet ≤ wall thickness

Examples `tests/alltag/fehler/f13_shell_verrundung.scad`

Cause The inner surface of a convex fillet with radius r < t degenerates when offset by t; SMLib only reports SM_ERR, the fallback via booleans fails as well.

Fix If both paths fail, the facade determines the smallest convex radius of curvature of the faces that are not opened (normal differences on a 7×7 grid per face) and names it: "wall thickness 2 is not smaller than the radius of a convex rounding of the body (about 2): the inner surface degenerates there (rounding radius must be > wall thickness)".

Evidence

`f13_shell_verrundung`: before V 59,848.6 mm³

1 Fehler: shell gescheitert (2 mm): SmApiShellBrepFull failed: SM_ERR (1001); shell from booleans: SmApiOffsetBrepFull failed: SM_ERR (1001) → after V 59,848.6 mm³

1 Fehler: shell gescheitert (2 mm): shell: wall thickness 2.000000 is not smaller than the radius of a convex rounding of the body (about 2.000000): the inner s

Tests: `f13_shell_thicker_than_a_rounding_names_the_radius` **ok**

F6, F11, F12, F15 – kernel bugs with message, without fix

Examples `tests/alltag/fehler/f06_fase_konkave_ecke.scad`, `tests/alltag/fehler/f11_stift_deckungsgleich.scad`, `tests/alltag/fehler/f12_shell_loft_quadrat_kreis.scad`, `tests/alltag/fehler/f15_knauf_kehle.scad`

Cause F6: SMLib has no working corner type for the corner "convex edge ends at a short concave edge" (SM_ERR; the chain is connected, filleting one after another does not help). F11: coincident cylinder surfaces (pin 3.0 in bore 3.0); offsets of 1e-6…1e-4 of the model size change nothing. F12: `SmApiShellBrepFull` of the loft square → circle (NULL_POINTER in SmMerge) and the fallback via booleans fail. F15: fillet sweep d 6 / sphere d 9 (SM_ERR) – with d 10 it works.

Fix No fix within the scope of this report; the messages mark the edges (F6, F15) or name the operation. Workarounds: zones on grid lines instead of on the plane (F6), 0.05 mm clearance (F11), difference of two lofts (F12), sphere d 10 (F15).

Evidence

`f06_fase_konkave_ecke`: before V 26,174.0 mm³

1 Fehler: Fase gescheitert (0.5 mm): chamfer failed: SM_ERR (1001) → after V 26,174.0 mm³

1 Fehler: Fase gescheitert (0.5 mm): chamfer failed: SM_ERR (1001)

`f11_stift_deckungsgleich`: before V 10,165.1 mm³

1 Fehler: Vereinigung gescheitert: boolean union failed: SM_ERR_NULL_POINTER (1006) → after V 10,165.1 mm³

1 Fehler: Vereinigung gescheitert: boolean union failed: SM_ERR_NULL_POINTER (1006)

`f12_shell_loft_quadrat_kreis`: before V 515,686.3 mm³

1 Fehler: shell gescheitert (2 mm): SmApiShellBrepFull failed: SM_ERR_NULL_POINTER (1006); kernel trace: SM_ERR_NULL_POINTER at SmMerge.cpp:672; shell from bool → after V 515,686.3 mm³

1 Fehler: shell gescheitert (2 mm): SmApiShellBrepFull failed: SM_ERR_NULL_POINTER (1006); kernel trace: SM_ERR_NULL_POINTER at SmMerge.cpp:672; shell from bool

`f15_knauf_kehle`: before V 1,118.4 mm³

1 Fehler: Verrundung gescheitert (1 mm): fillet failed: SM_ERR (1001) → after V 1,118.5 mm³

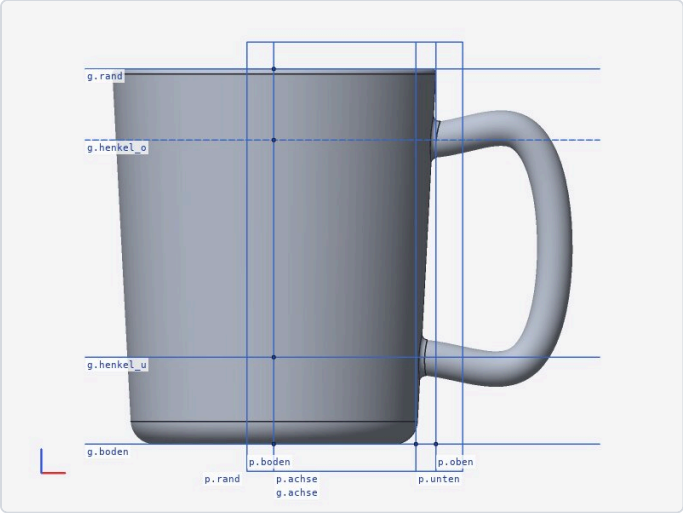
1 Fehler: Verrundung gescheitert (1 mm): fillet failed: SM_ERR (1001)

Tests: –

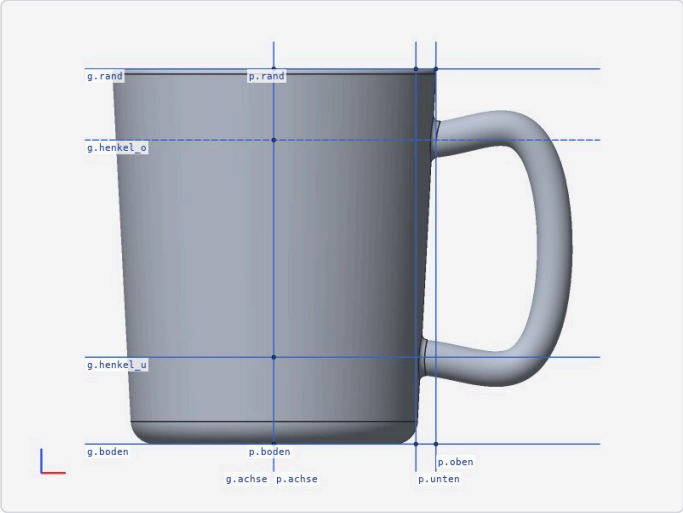
4 Display and CLI (S1–S4)

S1 – profile grid with rotate_extrude

Cause: a 2D grid (x = radius, y = height) is drawn in the coordinate system of its module, i.e. in the xy plane; after `rotate_extrude` the profile lies in the xz plane. **Fix (CLI):** `cad render --show grids` checks the references of each grid (`Scene.refs`): if all lie in 2D nodes below the same `rotate_extrude`, the grid is drawn in its profile plane (frame RE · Rx(90°) · RE^{-1} · N · frame_module) and a note is printed. Grids with references outside (or via variables) stay as they are.



Before: grid p in the xy plane, 95 mm behind the cup



After: grid p in the profile plane: p.unten at r 36, p.oben at r 41

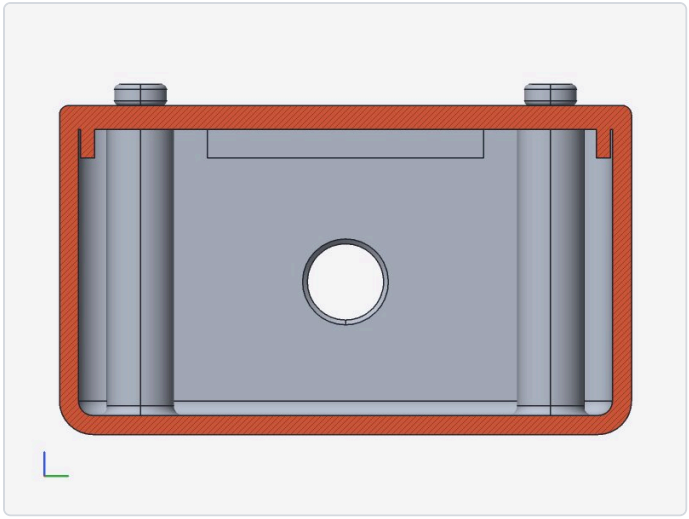
UI (not changed, `web/`): the 3D view draws grids from `Scene.grids[].frame`. For the same fix it would need either the rule above (references from `Scene.refs` → 2D nodes → `op_info.kind == "rotate_extrude"`) or, better, a frame that the scene supplies per grid (`Grid.display_frame`, computed in `cad-session` by the same rule). For dragging/snapping to grid points the handles would have to be mapped into the profile plane as well.

S3 – outline and visibility

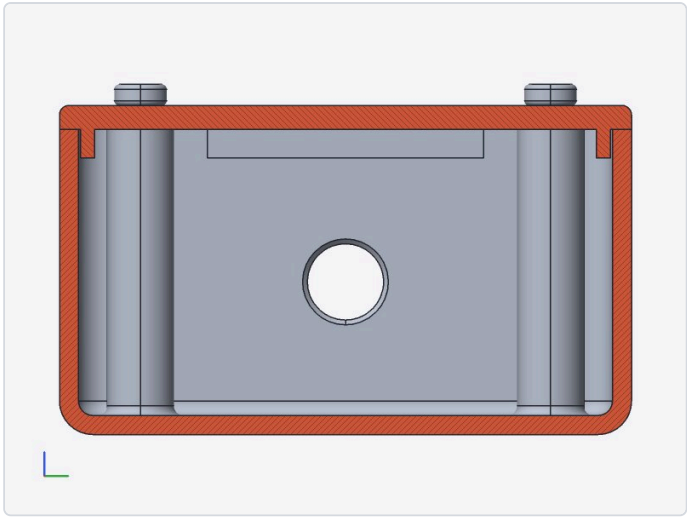
`cad outline` listed instances named in the body of a module (`dom[0]` … in `pi_platte`) under the file's „Benannte Instanzen“ (named instances) and their anchors as file anchors. Now: a separate section „Benannte Instanzen in Modulen (nur dort sichtbar)“ (named instances in modules, visible only there) grouped by module; in the JSON `scope: "file" | "module"` and `in_module`; anchors only of the file instances. That `measure --between dom[0].oben …` still works in file context is language semantics (cad-eval) and remains a proposal: resolve such names only via the instance chain (`platte.dom[0].oben`) or with a warning.

S4 – section caps and grid labels

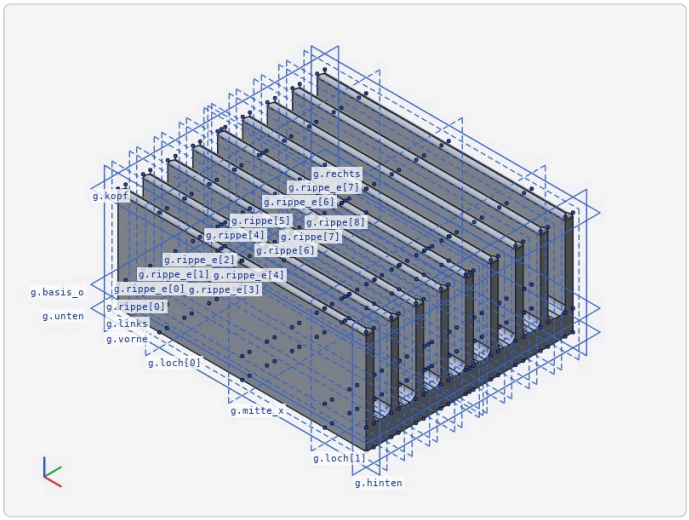
A model made of parts was drawn in section as one mesh of the root: the caps of touching parts merged. Now, as soon as the parts make up the whole model, it is drawn per part: its own cap with alternating hatching and its own section contour. Grid labels: rows of indexed planes (`rippe[0]` … `rippe[7]`) get one label `g.rippe[0..7]`; 3 px remain between labels (before, they could touch directly).



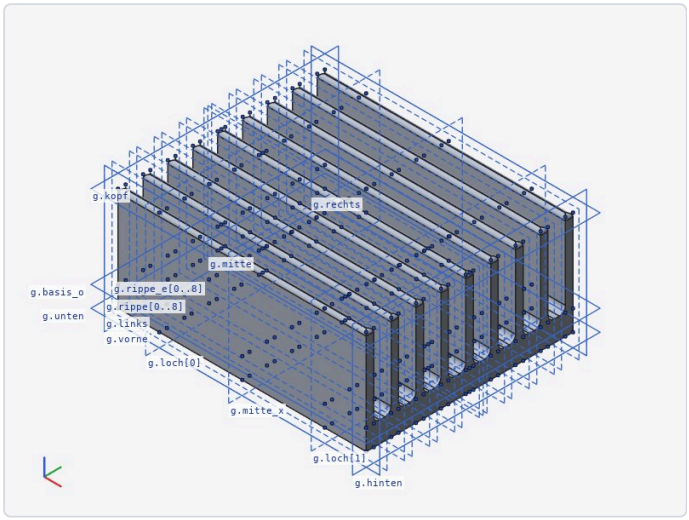
Before: lid and lower part one cap



After: lid separated (contour, opposite hatching)



Before: one label per rib



After: one label per row

S2 – plane names as values, anchor lists (assessment only)

- **Plane names as values** (`g[i % 2 == 0 ? loch_l : loch_r, …]`): the names in `g[…]` are selectors today, not expressions; the workaround via numbers (`g.loch_l`) loses the grid binding (snap, dragging). **Proposal:** a value type "plane" (`g.loch_l` still yields the number, but `plane(g.loch_l)` or a selector expression `g[sel(i % 2 == 0 ? "loch_l" : "loch_r"), …]`) that `g[…]` accepts; this keeps the origin for handles. Effort medium (cad-eval: selector evaluation, drag write-back only for literal selectors).
- **Anchor lists** (`anchor loch[i] = …` in a loop): fits the existing indexing of instances (`k[i]`) and planes (`rippe[i]`). **Proposal:** the same rule as for planes (name + index, outside `k.loch[2]`), snap targets per element. Effort low to medium.
- **Grid-local coordinates** for profiles in rotated planes (S1): a grid with its own frame (`grid p in xz { … }`) would also solve S1 in the language.

5 Run time

Finding: key rack 4.0 s kernel, but ≈ 16 s per evaluation. **Measurement:** stack samples with gdb (every ~0.8 s) during `cad check schluesselbrett.scad`:

```
vorher (Volumen 1e-8): 17 Stichproben (gdb, etwa alle 0,8 s zuzüglich Anhaltezeit)
   14    82 %  Bauteilvolumina nach dem Kernel-Lauf (sc_mass_props → SmApiBrepComputeVolume)
    3    18 %  Kernel-Lauf (SMLib, Stapel gekürzt)
nachher (Volumen 1e-6): 7 Stichproben (gdb, etwa alle 0,8 s zuzüglich Anhaltezeit)
    3    43 %  Kernel-Lauf: sc_boolean
    2    29 %  Bauteilvolumina nach dem Kernel-Lauf (sc_mass_props → SmApiBrepComputeVolume)
    1    14 %  Kernel-Lauf: sc_sweep
    1    14 %  sonst (Sprache, Szene, Bild)
```

Cause: after the kernel run the session computes the volume of each part (part list of the scene, `part_volumes`); after the mass properties (1e-4), `sc_mass_props` additionally called `SmApiBrepComputeVolume(…, 1e-8)`. The accuracy 1e-8 costs about 11 s for the swept hooks via adaptive Runge–Kutta integration per face (`SmFace::ComputePreciseProperties`). Measured relative volume error: 1e-6 → ≤ 4.4e-6, 1e-7 → ≤ 1.1e-7, 1e-8 → ≤ 1.5e-8 (sphere, torus, cone). **Fix:** 1e-6 (part volumes are displayed to 0.1 mm³; checks use 0.1 %).

cad render schluesselbrett.scad (median of 3)	before	after
Evaluation (evaluate_ms, without image)	16,669 ms	6,123 ms
of which kernel run (kernel_ms)	4,371 ms	4,153 ms
Image (render_ms)	996 ms	839 ms

The remainder between kernel and evaluation is now mainly the part volumes (1e-6) and tessellation/scene. A further step would be to compute the volumes only when they are needed (`outline`, `measure`, UI part list), or to take them from the mass properties (1e-4).

6 Open

- F2 (cylinder ∪ curved sweep), F3/F4 (sweeps of polygon profiles), F6 (corner type convex/concave), F11 (coincident cylinders), F12 (shell of a loft), F15 (fillet sweep/sphere): SMLib bugs; messages and workarounds as above.
- F10: the fillet across two faces is reported, not computed.
- F5, F8c, F9a have no correct result (collision, tangential contact); an automatic "overlap a little" would be possible, but changes the geometry without being asked to.
- S1 in the UI (proposal above), S2 and the visibility of `dom[0].oben` in file context (language semantics).
- The fillet check relies on coarse volumes (1e-2); errors that change the volume only slightly go undetected.

7 Test runs

New regression tests

`f09_examples_report_errors_instead_of_crashing` **ok**, `s1_profile_grid_of_rotate_extrude_is_drawn_in_the_profile_plane` **ok**, `s3_outline_lists_instances_named_in_modules_separately` **ok**, `s4_section_of_a_model_with_parts_draws_each_part` **ok**, `f01_symmetric_spline_profile_sweeps_along_a_curved_path` **ok**, `f03_open_polygon_sweep_names_the_cause` **ok**, `f01_off_centre_profile_follows_its_origin_on_a_curved_path` **ok**, `f07_translated_torus_selects_the_edge_on_its_core_circle` **ok**, `f02_failed_union_is_reported_once` **ok**, `f09_tangent_contact_under_union_fillet_fails_cleanly` **ok**, `f08_booleans_with_moved_tori_and_revolved_circles` **ok**, `f05_chamfer_that_leaves_an_open_shell_is_an_error` **ok**, `f13_shell_thicker_than_a_rounding_names_the_radius` **ok**, `f10_union_fillet_that_loses_the_shell_is_an_error` **ok**, `chamfer_result_must_be_closed` **ok**, `fillet_at_a_nearly_tangent_circle_fails_without_crash` **ok**, `moved_revolved_ring_unites` **ok**, `sweep_of_off_centre_and_spline_profiles_along_a_curve` **ok**, `tangent_torus_in_a_slab_is_an_error` **ok**, `volume_accuracy` **ok**, `difference_of_bodies_touching_along_an_edge_keeps_the_first` **ok**, `moved_rotated_and_scaled_tori_intersect` **ok**

Kernel: `crates/kernel-smlib/tests/alltag.rs`; engine: `crates/cad-engine/tests/alltag_fehler.rs` (reads the minimal examples); CLI: `crates/cad-cli/tests/geometrie.rs`.

Bug examples (check.sh --fehler)

```
Fehlerbeispiele (tests/alltag/fehler):
  behoben  f01_sweep_spline_profil.scad
  offen    f02_union_zylinder_sweep.scad: Fehler – gemeldet: Vereinigung gescheitert: boolean union failed:
SM_ERR_NULL_POINTER (1006)
  offen    f03_sweep_polygon16.scad: Fehler – gemeldet: sweep gescheitert: sweep: SMLib left 3 seam(s) between
the side faces of the polygon profile open (SMLib limit
  offen    f04_union_fillet_sweep_polygon.scad: Fehler – gemeldet: Verrundung gescheitert (1 mm): fillet failed:
SM_ERR (1001)
  behoben  f05_fase_volumen_null.scad (Fehler wie gewollt: Fase gescheitert (0.8 mm): chamfer failed: SMLib
returned a body that is not closed (faces missing): the blend)
  offen    f06_fase_konkave_ecke.scad: Fehler – gemeldet: Fase gescheitert (0.5 mm): chamfer failed: SM_ERR
(1001)
  behoben  f07_torus_zone.scad
  behoben  f08a_union_torus.scad
  behoben  f08b_union_ring_verloren.scad
  behoben  f08c_differenz_torus_tangential.scad (Fehler wie gewollt: Differenz gescheitert: the result is not a
closed solid: the operands touch tangentially or along an edge ther)
  behoben  f09a_absturz_tangentialer_dom.scad (Fehler wie gewollt: Vereinigung gescheitert: the result is not a
closed solid: the operands touch tangentially or along an edge th)
  behoben  f09b_absturz_zapfen_sweep.scad (Fehler wie gewollt: Verrundung gescheitert (1 mm): fillet failed:
SMLib cannot blend the corner where these edges meet (faces that)
  offen    f10_union_fillet_schale_verloren.scad: Fehler – gemeldet: Verrundung gescheitert (1 mm): fillet
failed: SMLib returned a body that is not closed (faces missing): the bl
  offen    f11_stift_deckungsgleich.scad: Fehler – gemeldet: Vereinigung gescheitert: boolean union failed:
SM_ERR_NULL_POINTER (1006)
  offen    f12_shell_loft_quadrat_kreis.scad: Fehler – gemeldet: shell gescheitert (2 mm): SmApiShellBrepFull
failed: SM_ERR_NULL_POINTER (1006); kernel trace: SM_ERR_NULL_POI
  behoben  f13_shell_verrundung.scad (Fehler wie gewollt: shell gescheitert (2 mm): shell: wall thickness
2.000000 is not smaller than the radius of a convex rounding o)
  offen    f15_knauf_kehle.scad: Fehler – gemeldet: Verrundung gescheitert (1 mm): fillet failed: SM_ERR (1001)
```

Everyday models, OpenSCAD comparison, STEP

```
19 Modelle, 0 mit Problemen, 112 s, Exporte in /tmp/alltag-9482ai1g (gelöscht)
compare.py: 33/33 files match
verify.py: 48/48 passed
```

cargo test, clippy, fmt, Wasm

```
cargo test --workspace: 426 passed, 0 failed
clippy -D warnings: sauber
fmt --check: sauber
node test-node.mjs: all node tests passed
```